ECONOMICS FOR ENGINEERS

Detailed Study Notes | Units 1 and 2

Syllabus-aligned explanations, formulas, worked examples, and exam guidance

Exam focus
HIGH PRIORITY labels mark topics repeated in the five past papers supplied. All syllabus headings are covered. Recurrence guides revision; it cannot predict the next examination.

Prepared for focused revision

In this guide

Unit 1: Foundations, scarcity and welfare, micro and macro, PPC, circular flow, demand and supply, elasticity, and equilibrium.

Unit 2: Utility, consumer equilibrium, indifference curves, budget constraints, and demand forecasting methods.

Exam toolkit: Formula sheet, diagrams, answer structures, and common mistakes.

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How to use this guide

Start with definitions and diagrams. Then practise starred formulas and examples without looking. For a 5-mark answer, give a definition, explain the principle, add a formula or labelled diagram, apply an example, and state the result.

UNIT 1 | Introduction and market basics

1. Economics: meaning, scope, and importance

Economics studies how individuals, firms, and societies make choices about using scarce resources to satisfy wants. Resources are limited while wants are many, so choices involve trade-offs and opportunity costs.

- Scarcity: resources cannot satisfy every want at once.
- Choice: selection among competing uses of resources.
- Opportunity cost: value of the next-best alternative given up when a choice is made.
- Incentive: a benefit or cost that influences a decision.

Example: A firm with a fixed budget chooses between a new machine and worker training. The opportunity cost of the choice is the benefit it gives up from the best alternative.

Economics helps consumers budget, firms plan production and costs, and governments assess policies affecting prices, employment, taxes, and public services.

2. Basic economic problems and resource constraints [HIGH PRIORITY]

Every society faces three allocation questions because resources are scarce:

- What to produce? Which goods and services, and in what quantities? Example: more public transport or more road capacity?
- How to produce? Which mix of labour, capital, land, and technology? Example: workers or automated machinery?
- For whom to produce? How should output and income be distributed? Example: who can afford housing when supply is limited?

Resource constraints limit inputs such as labour, land, capital, raw materials, time, and technology. A constraint forces prioritization. A hospital with a fixed budget and limited staff may have to allocate more resources to one service and fewer to another.

Welfare maximization means using limited resources to achieve the greatest possible social benefit. It involves productive efficiency (avoiding waste) and choices about who receives benefits. Total output can rise while gains are distributed unevenly, so efficiency and fairness may both matter.

Exam link: Connect constraints to choice, opportunity cost, and efficient allocation. Explain what is limited, what trade-off follows, and how allocation affects welfare.

3. Microeconomics and macroeconomics [HIGH PRIORITY]

Microeconomics studies individual decision-makers and specific markets. Macroeconomics studies economy-wide outcomes and aggregates.

- Micro: consumers, workers, firms, individual prices and markets. Example: how a price rise affects demand for coffee.
- Macro: national income, inflation, unemployment, economic growth. Example: how inflation affects purchasing power.
- Connection: higher national income (macro) may increase demand in particular product markets (micro).

Exam answer: State what each branch studies, compare their unit of analysis, and give one example of each.

4. Production Possibility Curve (PPC) [HIGH PRIORITY]

A Production Possibility Curve shows the maximum combinations of two goods an economy can produce with available resources and technology when resources are used efficiently.

- On the curve: productive efficiency.
- Inside: unemployed or inefficiently used resources.
- Outside: unattainable with current resources and technology.
- Outward shift: greater capacity from more resources, skills, or technology.
- Inward shift: lower capacity, for example after disaster or loss of capital.

The PPC illustrates scarcity because not every combination is attainable, and opportunity cost because more of one good generally requires less of another. Its slope shows how much of one good must be sacrificed. A bowed-out curve

reflects rising opportunity cost when resources are not equally suited to both uses.

Diagram: Put good X on the horizontal axis and good Y on the vertical axis. Draw a downward, bowed-out curve. Label a point on it efficient, one inside underused, and one outside unattainable.

Significance: Explain resource allocation, scarcity, opportunity cost, efficiency, unemployment of resources, and economic growth.

5. Circular flow of economic activities [HIGH PRIORITY]

The circular flow model shows how households and firms exchange resources, goods and services, and money. Households own factors of production; firms use them to produce output.

Factors of production: land (natural resources), labour (human effort), capital (produced tools and equipment), and enterprise (organization and risk-taking).

- Factor services flow from households to firms.
- Wages, rent, interest, and profit flow from firms to households.
- Goods and services flow from firms to households.
- Consumer expenditure flows from households to firms.

Diagram to reproduce: Households -- factors of production --> Firms; Firms -- wages, rent, interest, profit --> Households; Firms -- goods and services --> Households; Households -- spending --> Firms.

The exchange of resources and output is the real flow; payments and income are the money flow. A larger model adds government (taxes and spending), financial institutions (savings and investment), and the foreign sector (exports and imports).

Example: When a person works for a company, labour flows to the firm and wages flow back. The worker spends some income on goods and services.

Exam focus: Explain both real and money flows, draw and label the two-way exchanges, and give an example.

6. Demand and supply: foundations [HIGH PRIORITY]

Demand is the quantity consumers are willing and able to buy at different prices during a stated period, other things equal. The law of demand says price and quantity demanded usually move in opposite directions: price rises, quantity demanded falls; price falls, quantity demanded rises. The demand curve usually slopes downward. Substitution and income effects help explain this pattern.

Supply is the quantity producers are willing and able to sell at different prices during a stated period. The law of supply says price and quantity supplied usually move in the same direction. Higher prices can make additional production profitable; the supply curve usually slopes upward.

Movement vs. shift:

- A change in the good's own price causes movement along the existing curve.
- A change in another determinant shifts the entire curve.
- Say “quantity demanded/supplied changes” for movement; say “demand/supply changes” for a shift.

7. Factors affecting demand and supply [HIGH PRIORITY]

Demand shifters

- Income: higher income raises demand for a normal good; it may reduce demand for an inferior good.
- Tastes/preferences: greater preference raises demand (e.g., a health trend raises demand for low-sugar products).
- Related goods: higher price of a substitute may raise demand for this good; higher price of a complement may lower it. Tea and coffee can be substitutes; cars and fuel are complements.
- Expectations: expected future price increases may raise current demand.
- Number of buyers: more consumers generally raise market demand.

Supply shifters

- Input costs: higher costs reduce supply; lower costs increase it.
- Technology/productivity: improvement usually increases supply.
- Taxes/subsidies: a production tax tends to reduce supply; a subsidy tends to increase it.

- Number of sellers: more producers usually increase market supply.
- Expectations: expected future prices may change how much is sold now.
- Natural conditions/disruptions: favourable conditions raise supply; drought or transport disruption can reduce it.

Application: Explain which factor changes, identify which curve shifts and in which direction, then discuss effects on equilibrium.

8. Elasticity of demand and supply [HIGH PRIORITY]

Elasticity measures how responsive one variable is to a change in another. This is one of the most frequently examined topics.

Price elasticity of demand (PED)

PED = percentage change in quantity demanded / percentage change in price

PED is normally negative because price and quantity demanded move oppositely. Many answers interpret the absolute value.

- Absolute PED > 1: elastic; quantity changes proportionally more than price.
- Absolute PED < 1: inelastic; quantity changes proportionally less than price.
- Absolute PED = 1: unit elastic.
- PED = 0: perfectly inelastic in the idealized case.

Example: Price rises 10%, quantity demanded falls 20%. PED = -20% / 10% = -2; demand is elastic.

Factors: availability of substitutes (more substitutes -> more elastic); necessity vs. luxury (necessities often less elastic); budget share (larger share often more elastic); time to adjust (more time often means more elastic); how narrowly the product is defined.

Total revenue = price x quantity. With elastic demand, a price cut tends to raise total revenue. With inelastic demand, a price rise tends to raise total revenue. With unit elasticity, revenue is approximately unchanged, all else equal.

Income elasticity of demand (YED)

YED = percentage change in quantity demanded / percentage change in income

- Positive YED: normal good; demand rises as income rises.
- Negative YED: inferior good; demand falls as income rises.
- Normal good with YED above 1 is often called a luxury; between 0 and 1, a necessity. These are broad tendencies.

Cross-price elasticity of demand (XED)

XED of A with respect to B = % change in demand for A / % change in price of B

- Positive XED: goods generally behave as substitutes.
- Negative XED: goods generally behave as complements.
- Near zero: little relationship.

Example: if coffee price rises and demand for tea rises, the two behave as substitutes in that context.

Price elasticity of supply (PES)

PES = percentage change in quantity supplied / percentage change in price

PES is usually positive. Supply is elastic when quantity supplied changes proportionally more than price, and inelastic when it changes proportionally less. Spare capacity, accessible inputs, storability, and time to adjust tend to make supply more elastic. Fixed capacity, long production times, and scarce inputs tend to make it less elastic.

Percentage change = (new - original) / original x 100. For larger changes, if requested, use the midpoint method: (new - old) / average of new and old x 100.

Answer checklist: Define the elasticity; write the formula; substitute values; interpret sign and magnitude; give an example or factor.

9. Market equilibrium price [HIGH PRIORITY]

Market equilibrium is the price and quantity at which quantity demanded equals quantity supplied. There is no pressure from shortage or surplus to change price, assuming other conditions stay constant.

- Price below equilibrium: shortage (Qd > Qs); competition among buyers tends to push price up.
- Price at equilibrium: Qd = Qs; no shortage or surplus.
- Price above equilibrium: surplus (Qs > Qd); unsold stock encourages sellers to lower price.

Effects of shifts

- Demand increases, supply unchanged: equilibrium price and quantity generally rise.
- Demand decreases: both generally fall.
- Supply increases: price generally falls, quantity rises.
- Supply decreases: price generally rises, quantity falls.

Face-mask example: Concern about infection shifts demand right. If production is constrained, the initial supply is relatively fixed, creating a shortage and upward pressure on price. Firms may later expand capacity, shifting supply right. The final outcome depends on both shifts.

Diagram: Price on vertical axis, quantity on horizontal; downward D, upward S, intersection E. Show a shift with an arrow and label the new equilibrium.

UNIT 2 | Consumer choice and demand forecasting

10. Theory of utility [HIGH PRIORITY]

Utility is the satisfaction or benefit a consumer gets from goods and services. It describes preferences and is not necessarily directly measurable.

- Total utility (TU): total satisfaction from a quantity consumed.
- Marginal utility (MU): extra satisfaction from one more unit.
- MU = change in total utility / change in quantity.

Law of diminishing marginal utility: As a person consumes more units of a good in a given period, each additional unit often adds less satisfaction, other things equal. TU rises while MU is positive, reaches a maximum when MU is zero, and can fall if MU becomes negative.

Example: the first glass of water when thirsty may have high MU; another after thirst is satisfied adds little.

11. Consumer equilibrium using utility [HIGH PRIORITY]

A consumer chooses an affordable bundle that gives the greatest satisfaction. For goods X and Y, an interior optimum follows the equi-marginal principle:

$$MUx / Px = MUy / Py$$

If MUx/Px exceeds MUy/Py, shifting some spending toward X may raise satisfaction. The consumer reallocates spending until the ratios are equal or a constraint applies. The budget must also hold: PxQx + PyQy = income when all income is spent.

This condition assumes comparable preferences, known prices, and a feasible bundle. Corner choices may occur if the best choice involves spending on only one good.

12. Indifference curve analysis [HIGH PRIORITY]

An indifference curve (IC) shows combinations of two goods giving the same satisfaction.

- Downward sloping: more of one good requires less of the other to keep satisfaction unchanged.
- Higher curves are preferred, assuming both goods are desirable.
- Usually convex to the origin, reflecting diminishing marginal rate of substitution.
- Curves do not cross, because crossing implies inconsistent rankings.

Marginal rate of substitution (MRS) is the amount of Y a consumer is willing to give up for one extra X while remaining equally satisfied. It corresponds to the absolute slope of the IC.

Plotting from a table: If bundles are A(1 tea,12 coffee), B(2,8), C(3,5), D(4,3), E(5,2), put tea on the horizontal axis and coffee on the vertical. Plot and label the points, then draw a smooth downward curve. If the question states these bundles give equal satisfaction, all lie on the same IC.

13. Budget constraints and consumer optimum [HIGH PRIORITY]

A budget constraint shows what the consumer can afford given income and prices.

Affordable bundles: PxQx + PyQy <= M

Budget line: PxQx + PyQy = M

If all income is spent on X, Qx = M/Px; on Y, Qy = M/Py. With Qx horizontal and Qy vertical, slope = -Px/Py.

- Income rises with prices unchanged: parallel outward shift.
- Income falls: parallel inward shift.
- Price of X falls: X intercept rises; budget line pivots outward around the Y intercept.
- Price of X rises: X intercept falls; budget line pivots inward.

Consumer equilibrium with ICs: The consumer chooses the highest attainable IC subject to the budget. At a typical interior optimum the budget line is tangent to the IC:

MRSxy = Px / Py

This means the consumer's willingness to trade goods matches the market trade-off. For differentiable utility at an interior optimum, this corresponds to MUx/Px = MUy/Py.

Key link: Budget constraint identifies affordable bundles; indifference curves rank bundles. Equilibrium is the most preferred affordable bundle.

14. Demand forecasting: purpose and process [HIGH PRIORITY]

Demand forecasting systematically estimates future demand. It supports production, inventory, capacity, staffing, purchasing, pricing, and budgeting.

Basic workflow: 1. Define product, market, and forecast horizon. 2. Collect reliable historical data and relevant factors (price, income, seasonality, promotions). 3. Choose a method and state assumptions. 4. Calculate and compare the forecast with actual demand. 5. Revise assumptions or method when conditions change.

Forecasts are uncertain. Poor data, unexpected events, changing preferences, competitors, and structural changes reduce accuracy. Long-range forecasts are often more uncertain than short-range forecasts.

15. Time-series forecasting [HIGH PRIORITY]

A time series is a sequence of observations over time (e.g., monthly sales). Forecasting uses historical patterns to estimate future values.

- Trend: long-run upward or downward direction.
- Seasonal variation: regular repeating pattern (e.g., heater demand in winter).
- Cyclical variation: longer waves associated with economy-wide conditions.
- Irregular variation: unpredictable event such as strike or disruption.

Simple linear trend: Yt = a + bt, where Yt is demand at time t, a the baseline, and b the estimated change per period. Positive b means growth; negative b means decline. Seasonal patterns may need separate adjustment.

Strength: straightforward when historical patterns are stable. Limitation: assumes the past remains informative; structural change and unusual events reduce accuracy.

16. Moving averages [HIGH PRIORITY]

A moving average smooths short-term fluctuations by averaging a fixed number of recent periods. When a new observation arrives, drop the oldest and include the new one.

n-period moving average = sum of latest n demand observations / n

Example: Demand is 100, 120, and 110 units over three months. Next-month forecast = (100 + 120 + 110) / 3 = 110 units.

A shorter window reacts faster but is less smooth. A longer window smooths more but reacts slowly to a real trend change. It is easy to calculate but does not explain why demand changed.

Exam calculation: Show observations included, sum, divide by window size, and state forecast with units and period.

17. Exponential smoothing [HIGH PRIORITY]

Exponential smoothing combines latest actual demand with the previous forecast, weighting recent information.

Ft+1 = alpha At + (1 - alpha) Ft

Ft+1 = next forecast; At = current actual demand; Ft = current forecast; alpha is between 0 and 1.

Example: At = 120, Ft = 100, alpha = 0.30: next forecast = 0.30(120) + 0.70(100) = 36 + 70 = 106 units.

Higher alpha gives more weight to the latest actual and responds faster. Lower alpha gives more weight to the previous forecast and produces a smoother, slower response. Basic smoothing does not separately model trend or seasonality.

18. Regression technique [HIGH PRIORITY]

Regression analysis estimates the relationship between demand and explanatory variables such as price, income, advertising, or population. Future values of the factors can then be used to forecast demand.

Simple regression: Q = a + bX

Q = demand; X = explanatory variable; a = intercept; b = estimated change in demand for a one-unit change in X. Demand-price coefficient is often negative; demand-income for a normal good is often positive.

Multiple regression: Q = a + b1P + b2Y + b3A P may be price, Y income, and A advertising. Each coefficient estimates the relationship with demand while holding other included factors constant.

Steps: choose demand and plausible explanatory variables; collect matching observations; estimate the equation; interpret signs and units; insert expected future values; compare forecasts with actual outcomes.

Example: An advertising coefficient of 4 associates one additional unit of advertising expenditure with 4 additional demand units, holding included factors constant. Interpretation depends on units.

Regression can quantify associations and support scenarios, but needs suitable data. Omitted factors, poor measurement, or changing relationships can mislead. Association alone does not prove causation.

EXAM TOOLKIT | Formulas and answer plans

Formula sheet

- PED = % change in quantity demanded / % change in price.
- YED = % change in quantity demanded / % change in income.
- XED (A with respect to B) = % change in demand for A / % change in price of B.
- PES = % change in quantity supplied / % change in price.
- Percentage change = (new - original) / original x 100.
- Total revenue = price x quantity.
- MU = change in TU / change in quantity.
- Utility equilibrium: MUx/Px = MUy/Py.
- Budget constraint: PxQx + PyQy <= M.
- Budget line slope: -Px/Py.
- IC equilibrium: MRSxy = Px/Py.
- Moving average = sum of window observations / number of periods.
- Exponential smoothing: Ft+1 = alpha At + (1 - alpha) Ft.
- Linear trend: Yt = a + bt.
- Simple regression: Q = a + bX.

Diagrams to practise

- Circular flow: households, firms, factor services, factor income, goods/services, spending; real and money flows.
- PPC: goods on axes; efficient, underused, unattainable points; opportunity cost.
- Demand and supply: price and quantity axes, D, S, equilibrium; shortage/surplus or shift.
- Indifference curve and budget line: two goods, IC, budget line, highest attainable IC and tangency.

Reliable answer structures

Definition/short note: precise definition, distinguishing feature, example.

Five-mark explanation: definition; mechanism; formula or diagram; example; implication for price, quantity, choice, welfare, or forecast.

Market case: initial situation; identify curve shift and reason; diagram; new equilibrium price and quantity; short-run constraint or long-run response.

Forecast calculation: method and formula; data used; calculation; forecast with period and units; assumption or limitation.

Common mistakes

- Calling own-price change a shift; it causes movement along a curve.
- Confusing demand with quantity demanded, or supply with quantity supplied.
- Giving an elasticity value without interpreting sign and magnitude.
- Forgetting positive XED usually means substitutes, negative means complements.
- Using the wrong moving-average window or failing to drop its oldest value.
- Swapping actual demand and previous forecast in exponential smoothing.
- Drawing diagrams without axes, curve labels, or equilibrium.
- Explaining consumer equilibrium without showing affordability.

Final revision order: Start with elasticity, circular flow, utility/consumer equilibrium, and forecasting methods. Then practise demand/supply shifts, resource constraints and PPC, micro vs. macro, budget constraints, and indifference curves. Review every remaining syllabus heading at least once.